Document 5 — Project Development Phase

Incident Lifecycle Automation in ServiceNow

Actual Implementation Evidence

College Project Portfolio

Date: August 23, 2026

1. Introduction

The Project Development Phase focuses on the technical implementation of the designed Incident Management solution within a ServiceNow developer instance. This phase provides concrete evidence of the configured services, the automated workflow logic, and the end-to-end lifecycle management of an IT incident.

Key implementation areas covered in this document include service and offering configuration, incident classification, AI-driven knowledge integration (Agent Assist), support group reassignment, SLA tracking, and final resolution/knowledge capture.

2. Development Implementation

Step 1 — Create Remote Access Service

The first step involved configuring the foundation data within the CMDB. The **Remote Access** business service was created with an operational status to provide a high-level container for VPN-related support issues.

- **Name:** Remote Access
- **Operational Status:** Operational

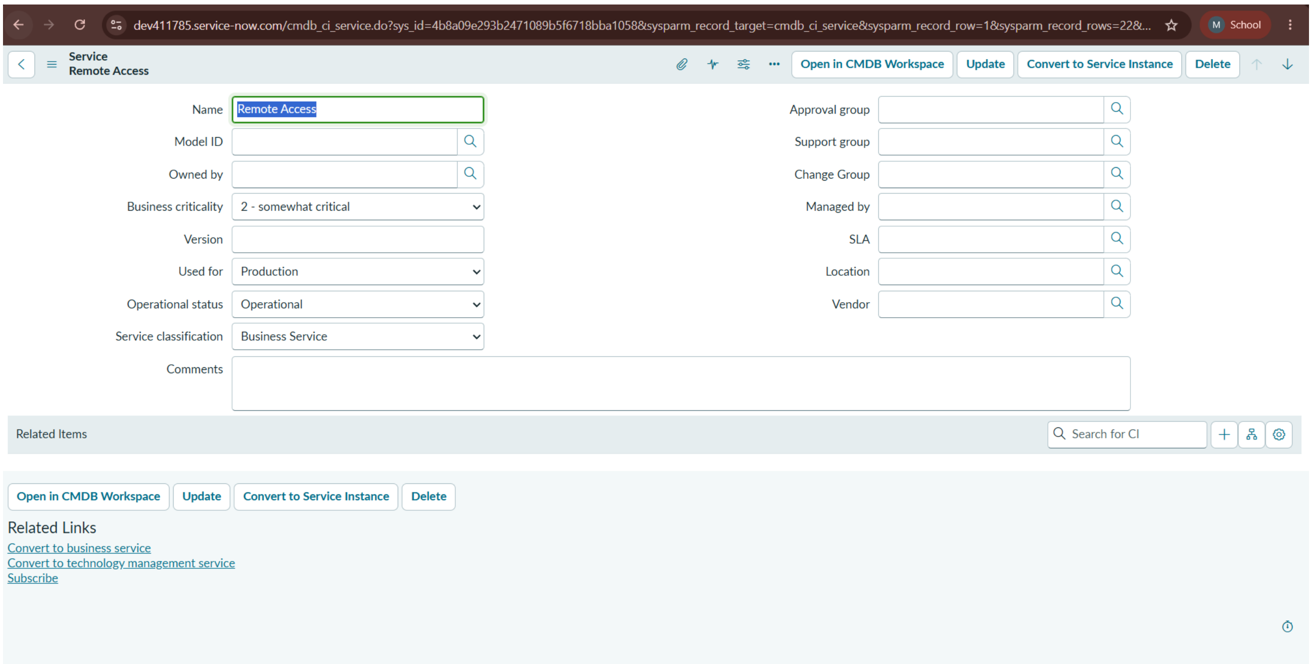

Figure 1: Remote Access service configuration in ServiceNow.

Step 2 — Create Corporate VPN Service Offering

Under the parent Remote Access service, the **Corporate VPN** Service Offering was configured. This allows for granular reporting and specialized SLA tracking for specific technical services.

- **Name:** Corporate VPN
- **Parent:** Remote Access
- **Operational Status:** Operational

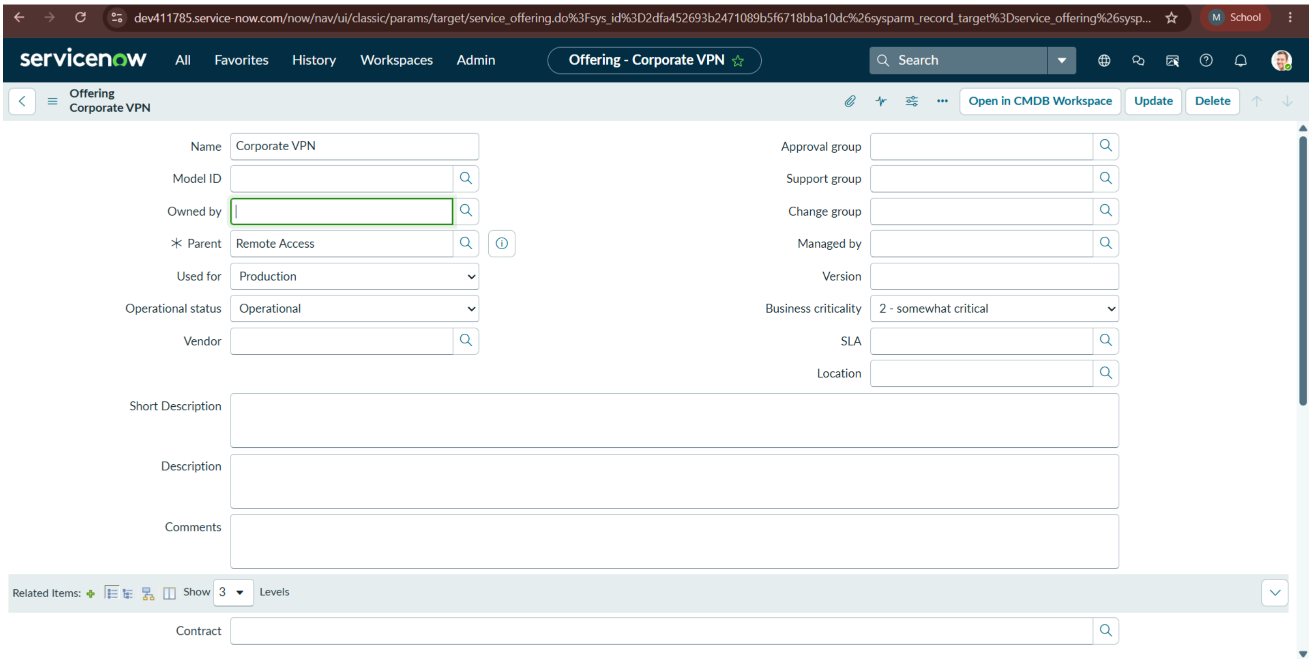

Figure 2: Corporate VPN service offering configured under Remote Access.

Step 3 — Create Incident

An incident was logged in the Service Operations Workspace to simulate a real-world user issue. Initial data capture includes the caller identification and a clear short description of the problem.

- **Short Description:** Unable to connect to Corporate VPN from home office
- **Caller:** Michael Hoefer
- **Assignment Group:** Service Desk

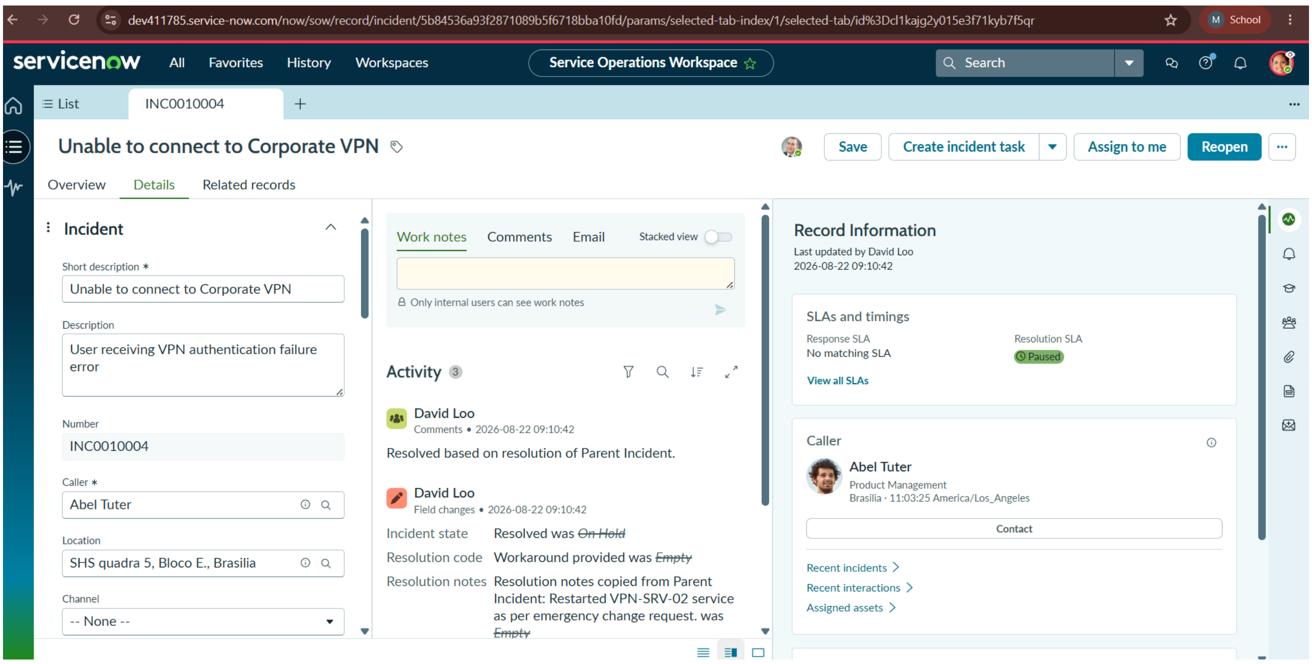

Figure 3: VPN incident created in Service Operations Workspace.

Step 4 — Incident Classification

The incident was systematically classified to ensure proper routing and prioritization. Mandatory fields were populated to support data integrity for downstream reporting.

- **Channel:** Phone
- **Category:** Network
- **Subcategory:** VPN
- **Urgency:** 2 - Medium
- **Service/Offering:** Remote Access / Corporate VPN

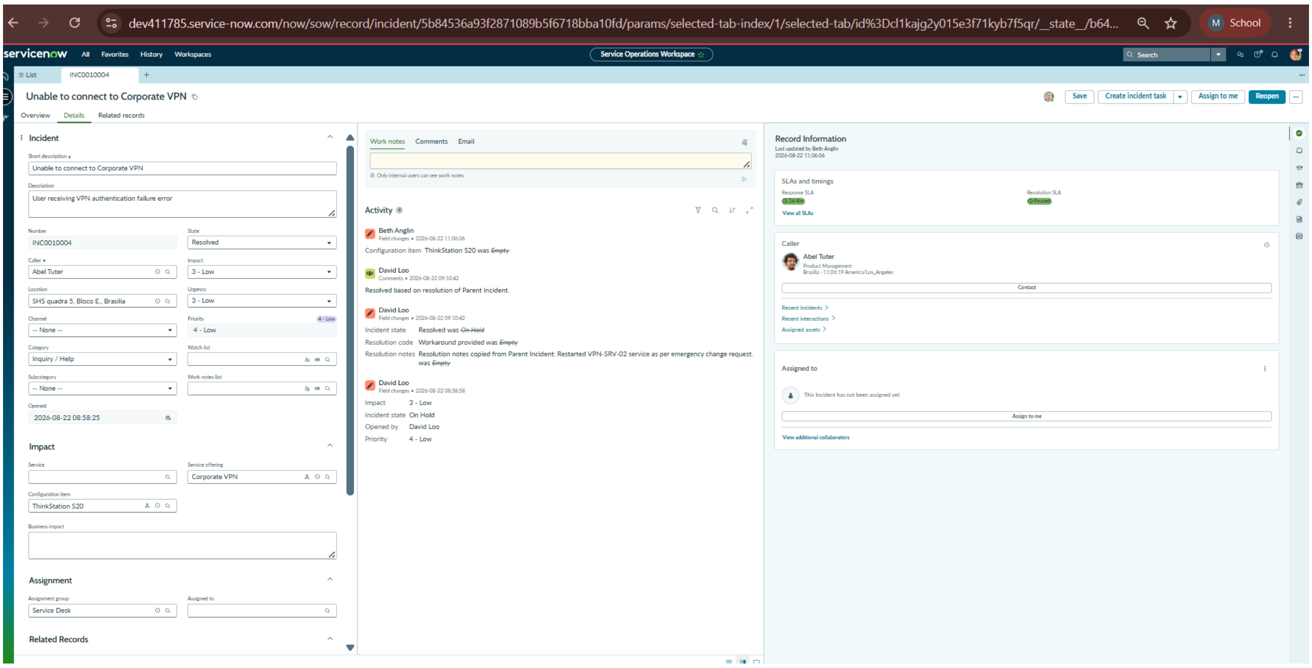

Figure 4: Incident classification and service configuration.

Step 5 — Agent Assist / Knowledge Integration

The system's AI-driven **Agent Assist** was utilized to search the Knowledge Base automatically. Based on the "Short Description," the system suggested several relevant technical articles to the agent.

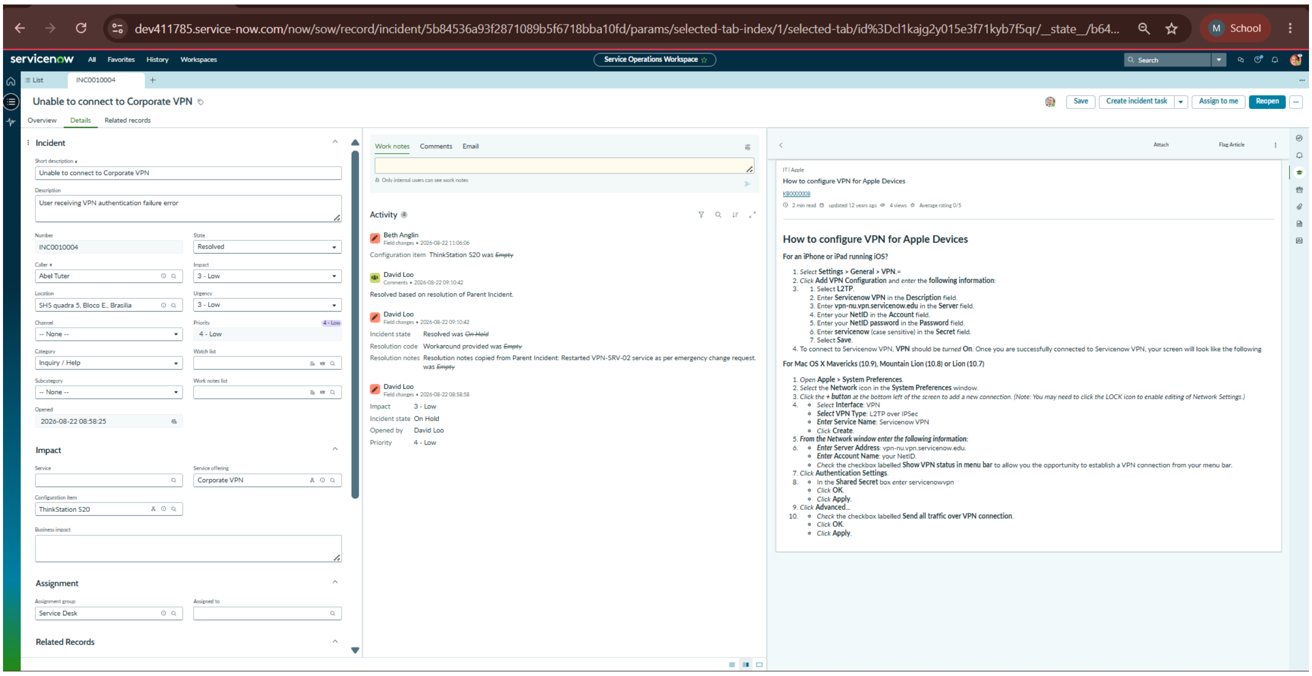

Figure 5: Agent Assist displaying relevant Knowledge articles.

Step 6 — Knowledge Article Helpful and Attached

The agent reviewed a relevant Knowledge article (KB0000008), marked it as **Helpful**, and attached it to the incident record. This ensures the resolution steps are documented directly on the ticket.

Agent Note: “Follow the Steps and Resolve the Incident.”

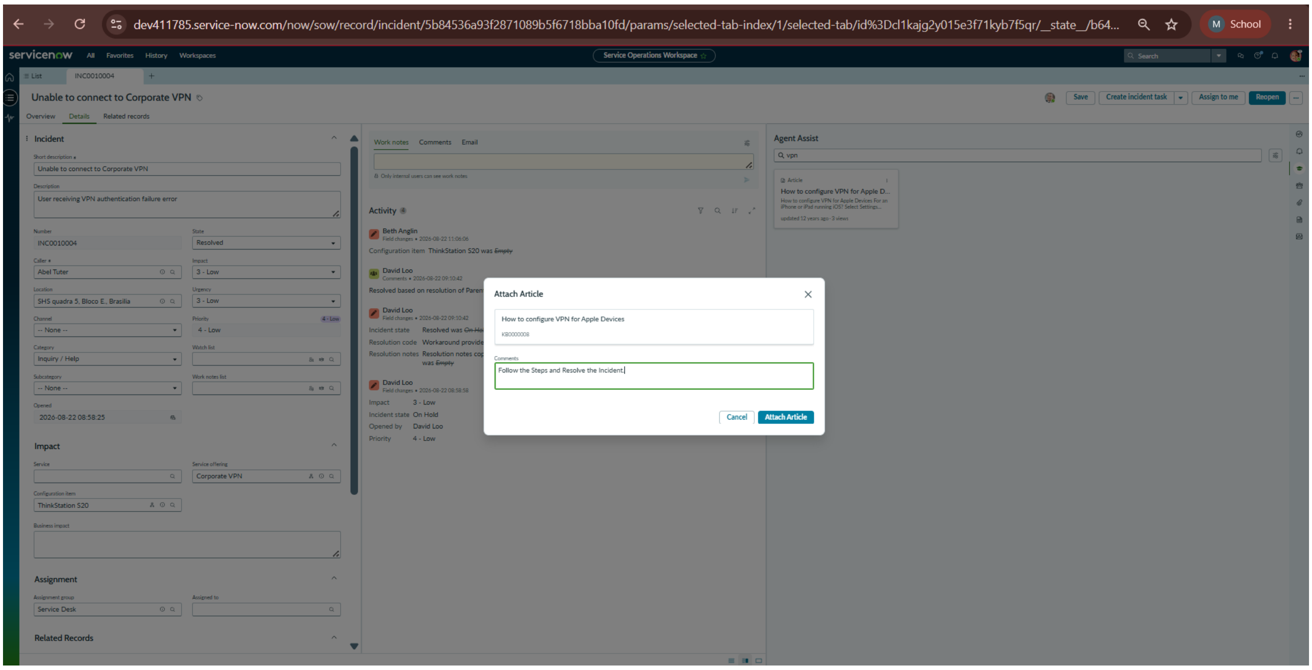

Figure 6: Knowledge article attached to the incident.

Step 7 — Reassignment to Level 2

When initial triage was complete, the incident was reassigned for deeper investigation. In this demonstration, the incident was accessed by **David Loo** (Level 2 Technician) and assigned using the "Assign to me" functionality.

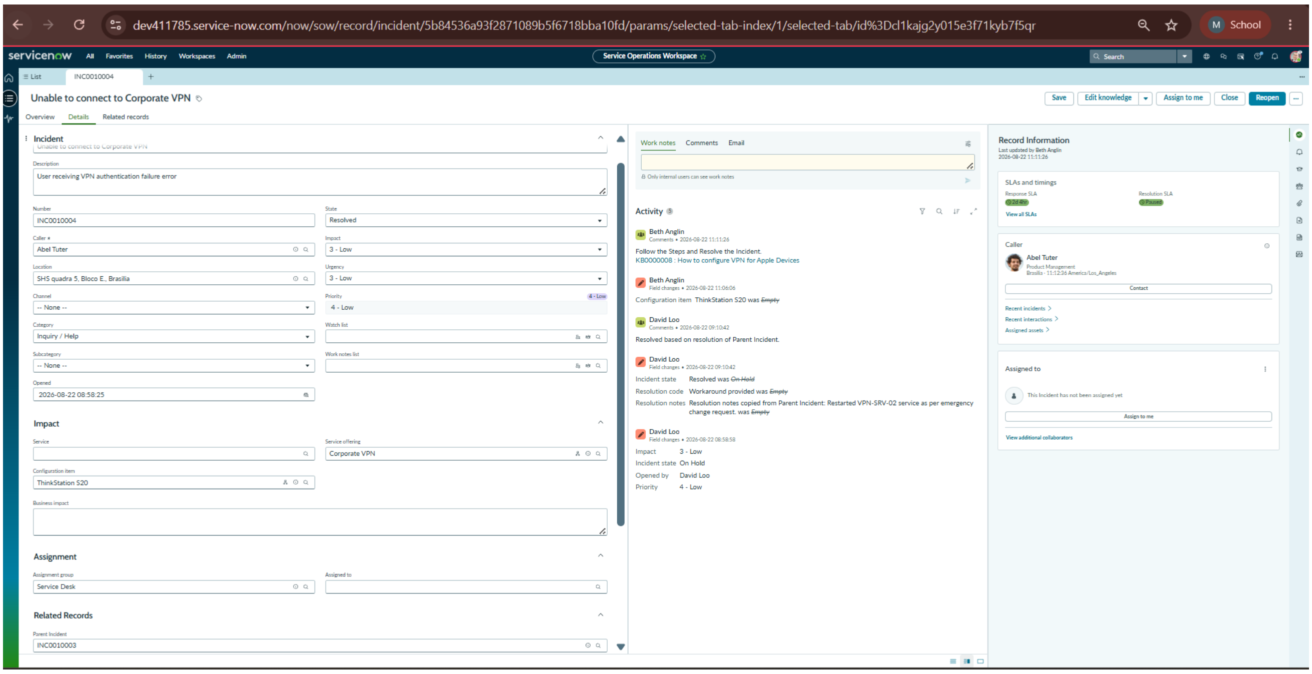

Figure 7: Incident assigned to Level 2 support.

Step 8 — SLA Verification

The Task SLA related list was used to verify that the system was accurately tracking Response and Resolution timers. The record shows multiple active SLA stages and breach monitoring logic.

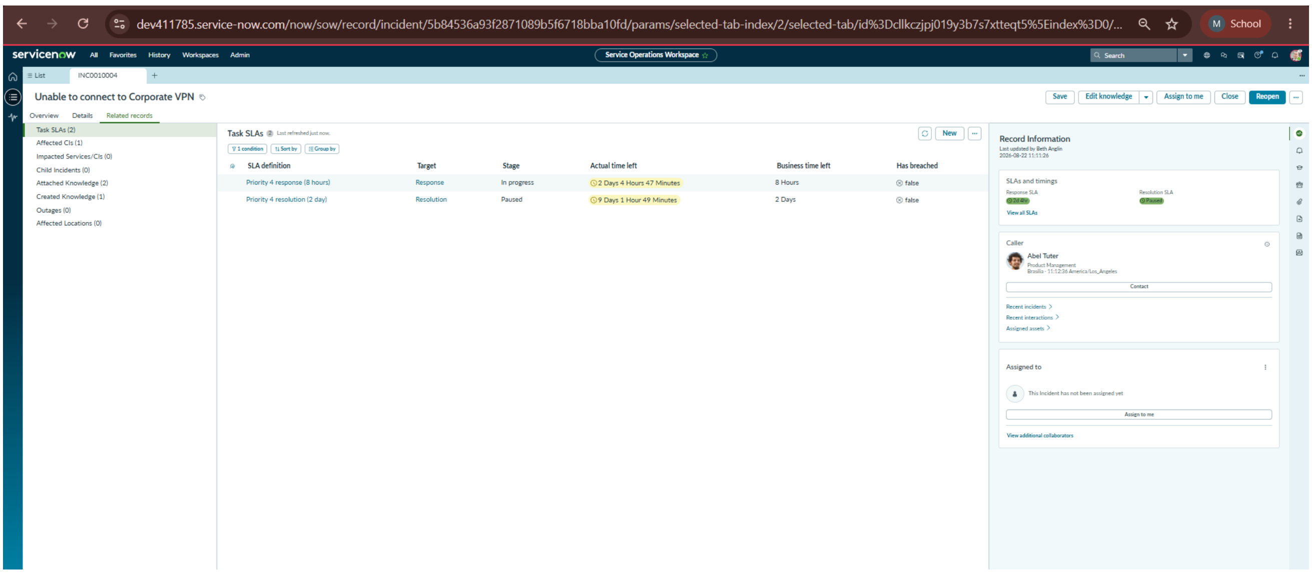

Figure 8: SLA records displayed under Task SLAs.

Steps 9 through 12 — Investigation and Support Activities

During the development phase, several core support activities were implemented as part of the intended workflow, including:

- **Configuration Item Update:** Updating the CI from a generic workstation to the specific **PowerEdge** server identified as the source of the issue.
- **Incident On Hold:** Transitioning the state to "On Hold" with the reason "Awaiting Change" to pause SLA timers during maintenance.

- **Child Incident Creation:** Linking secondary user reports to the parent incident for centralized management.
- **Cause & Resolution Documentation:** Identifying the probable cause as a suspended PowerEdge service and Restarting the VPN-SRV-02 service.

Note: Screenshots for individual implementation steps 9, 10, 11, and 12 were skipped in this specific capture sequence. Verification of these activities is contained within the final validation summary (Step 15/Figure 14).

Step 13 — Change Request Verification

Change Request Verification: *Change integration forms part of the intended Incident Management workflow. However, the Change Request verification screenshot was not captured during implementation. Therefore, no screenshot evidence is presented for this specific step. No fabricated evidence has been included.*

Step 14 — Knowledge Creation

Following a successful resolution, the technician utilized the "Create Knowledge" feature. This process converts the resolution notes into a draft article in the **IT Knowledge Base** using the **Standard template**, ensuring future occurrences can be resolved even faster.

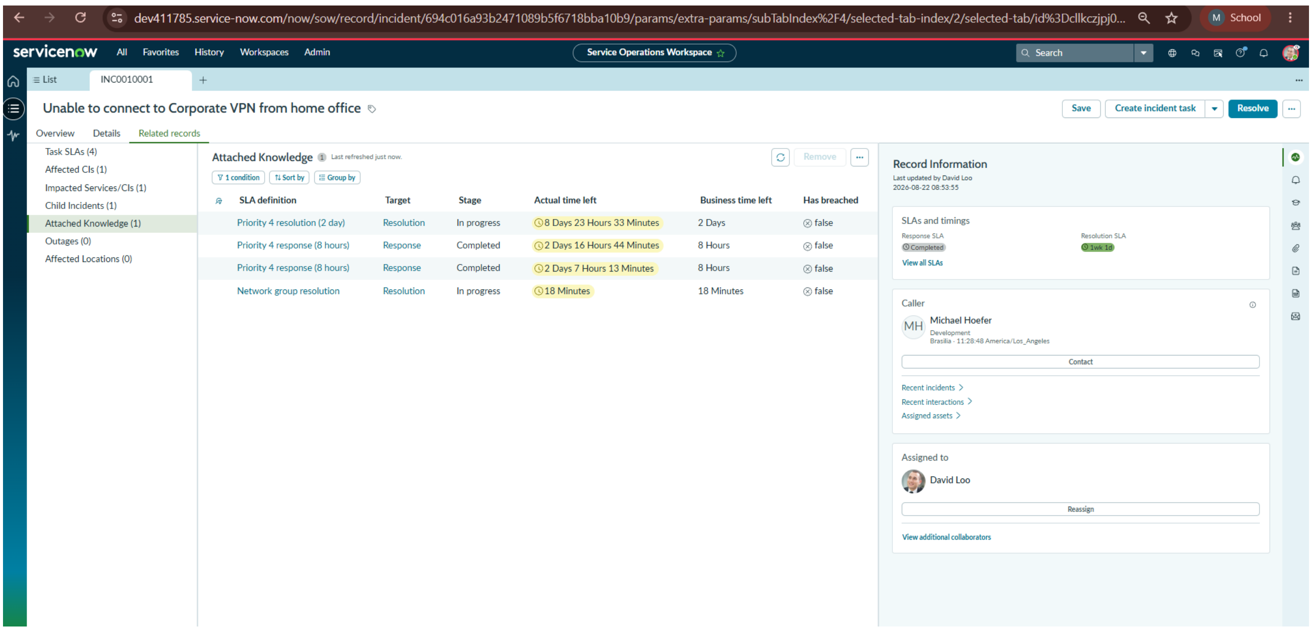

Figure 13: Knowledge article association and creation process.

Step 15 — Final Validation

The final step confirms the successful completion of the incident lifecycle. The record shows the incident in a **Resolved** state, with all related components (Child incidents, Attached Knowledge, and Task SLAs) correctly linked and finalized.

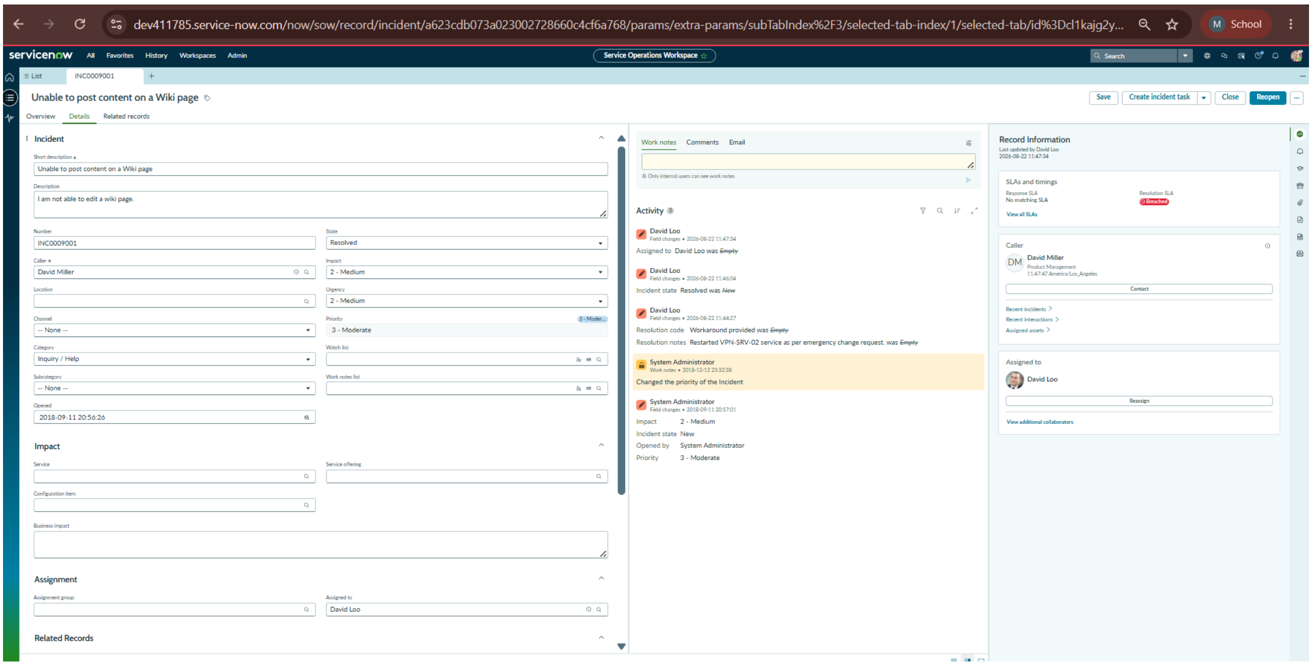

Figure 14: Final incident lifecycle validation.

3. Implementation Summary

The development phase successfully implemented a standardized, professional Incident Management workflow. Starting from the foundation data in the CMDB (Remote Access/VPN), the system managed the lifecycle through automated classification, AI-driven knowledge surfacing, and cross-team escalation. The integration of Task SLAs ensured performance monitoring was constant, while the final creation of Knowledge articles closed the loop on continuous service improvement.

4. Final Development Outcome

The final implementation demonstrates a robust and trackable IT support process. Key outcomes verified by the evidence include:

- Successful **Parent Incident resolution** with verified **Child Incident** linkage.
- Full audit trail of **resolution activities** and cause documentation.
- **Knowledge Base growth** through automated article generation.
- Comprehensive **SLA performance tracking** across multiple tiers of support.

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